

Alzheimer's Disease Classification Using CNN Optimized with the Adam Optimizer

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Abstract—This project aims to implement a Convolutional Neural Network (CNN) and optimized with adam optimizer for classification of the stages of Alzheimer's disease using MRI of the brain. The study uses the Best Alzheimer MRI Dataset (99% Accuracy), consists of four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The dataset was preprocessed by resizing, normalization, and batching to create consistent input for training. The training data was then inputted into a custom-built CNN architecture consisting of several convolutional and pooling layers designed to automatically learn spatial and structural features of brain MRI scans. The model's learning process was optimized using the Adam optimizer and a categorical cross-entropy loss function. This project highlights the power of CNN-based deep learning and its ability to classify the stages of Alzheimer's disease using MRI of the brain, furthering research within computer-aided diagnosis in medicine.

Index Terms—Alzheimer's Disease, Convolutional Neural Network (CNN), MRI Brain Images, Deep Learning, Image Classification, Medical Image Processing, Adam Optimizer

I. INTRODUCTION

Alzheimer's disease (AD) is a long-lasting, progressive neurodegenerative disorder that leads to irreversible death of brain cells and successive impairment on memory, reasoning and cognitive functions. It is the most common type of dementia; accounting for approximately 60-70% of all cases of dementia worldwide. The onset of the disease is gradual from a person starting with mild memory loss to becoming severely cognitively impaired, and losing independence. With an increasingly aging global community, the incidence of Alzheimer's is increasing exponentially, and is among the most pressing health and social difficulties of the 21st Century.

Although traditional image processing methods are useful, they normally require hand-crafted feature extraction, which is a manual process that must be specified by a human expert before using them, for whatever features of interest (such as texture, shape, or intensity). The manual process of developing the image descriptors limits scalability and may lack the ability

to account for complex structural changes that occur in the brain during the progression of Alzheimer's disease. Thus, the need has arisen for automated, intelligent diagnostic systems that can review brain images consistently, efficiently, and with minimal human bias.

Deep learning, a branch of Artificial Intelligence, has emerged as a powerful approach in the medical analysis of structural brain images. In many deep learning approaches, with Convolution Neural Networks (CNNs) being a specific type of architecture, the capacity for automatically learning the hierarchical predicted spatial feature representations from imaging data is exceptional. CNNs do not necessitate the manual feature extraction that traditional image processing requires; instead, CNNs learn the discriminative features, or descriptors, directly from raw images, which is invaluable in recognition problems, such as disease classification and detection. CNNs have successfully illustrated their potential in medical studies of cancer detection and classification, diabetic retinal disease detection and analysis, and tumor classification, highlighting their potential medical applications.

Using CNN with adam optimizer in Alzheimer's disease research facilitates the automated detection of subtle structural alterations in the brain, such as tissue atrophy, brain volume loss, and cortical thinning, which are known indicators of disease progression. By utilizing deep learning models to analyze brain MRI scans, the models outperform clinical expertise regarding the reliable classification of patients into Alzheimer's disease phases while reducing the time to diagnosis. Moreover, CNN-based models are applicable to a space in this domain known as computer-aided diagnosis (CAD)—an innovative use of decision-making with the objective of supporting clinician decision-making in a validated manner that yields quicker, more objective, and more reliable clinical decisions. This project seeks to explore the benefits and capabilities of CNNs to classify the range of an individual's Alzheimer's disease progression through MRI brain image

assessment. The primary aim is to investigate the effectiveness of convolutional architectures and the recognition of the structural and spatial characteristics of the brain demonstrating dementia. Ultimately, this project seeks to progress the development of intelligent, and effective systems for the purposes of supporting healthcare practitioners in the identification of Alzheimer’s disease and the classification of the disease, thereby assisting with management and treatment approaches for individual patients.

II. LITERATURE REVIEW

Research efforts focused on automated detection of Alzheimer’s disease (AD) from neuroimaging (MRI/PET) data, in early and mid-2020s, have primarily utilized convolutional neural networks (CNNs) and their modern variants, as we have seen a shift in the field from CNNs with fewer layers toward deeper architectures, transfer learning, and more ensemble/hybrid approaches. Several efforts [1][2][8][10] employ Inception-style or custom CNNs and report high classification performance, often following extensive preprocessing and augmentation. The classic deeper residual networks (ResNet family) are popular for their strengths in feature extraction [4][6][9], while efficient architectures such as EfficientNet [7] and ensembles of MobileNet [4] are used to balance classification accuracy and computational expense. More recent studies explore novel paradigms: self-supervised + ConvNeXt[13], ensembles of Vision Transformer and EfficientNet [14], hybrid approaches (CNN + machine learning classifier in [12] or CNN + feature of selection methods MICRO + MLP feature classifiers.

Datasets used across these studies were largely publicly available neuroimaging collections ADNI and OASIS, with the most common and including Kaggle MRI compilations and in some case custom-aggregated datasets [2][9]. Several efforts utilize MRI + PET or neuropsych measures alongside their MRI data to improve robustness [13][15]. Reported accuracy varied significantly across studies: several deep models and ensembles report high performance in the mid-90s or higher [1][2][4][7]. Some studies, meanwhile, claim to peak between 95-99% or higher in accuracy, [9] reports extremely high. Studies reporting results on a more moderate level of accuracy [3], do reflect different compositions of datasets including class balance, preprocessing, and evaluation.

Common limitations to accuracy, however, are still prevalent across the studies. Dataset size, class imbalance, and single site or homogeneous cohorts are discussed frequently in terms of reduced generalizability and/or potential increased overfitting [2][3][5]. High computational cost, specifically requiring powerful GPUs exists for very deep or high ensemble models [4][6][14]. Several papers are described as review or survey studies [5][11], these papers summarize approaches rather than develop new models with new validations, therefore collectively broadly applicable, reproducible models with clinical validations are limited. Interpretability is not discussed in depth and hybrid methods (e.g. CNN + SVM and MIC + SMOTE) report attempts to combine classical ML robustness

with deep feature extraction and generalization, but still report overfitting and dependency on dataset [12][15].

There are methodological trends that lead to better reported performance which are the same across domains: transfer learning from backbones pretrained on ImageNet, some sort of preprocessing like normalization and augmentation, ensembling the results from different architectures, and class balancing methods such as the use of augmentation or the combination of SMOTE. There is promise in exploring self-supervised learning, and also in combinations of MRI + PET or imaging + clinical features, for better robustness and capture of disease heterogeneity [13][15]. However, cross-study comparisons remain awkward and sometimes inconsistent due to factors such as train/test split, number of classes (binary vs multi-class staging), and evaluations. All of which will drive some very high accuracy claims that won’t reproduce in an external cohort.

III. METHODOLOGY

The methodology being proposed utilizes a Convolutional Neural Network (CNN) with adam optimizer model to classify stages of Alzheimer’s disease based on MRI brain images. The methodology consists of preprocessing steps (image resizing, normalization, and batching) to improve the training efficiency and accuracy. The trained CNN model extracts deep spatial features automatically to allow accurate and early identification of stages of Alzheimer’s.

A. Dataset overview

The Best Alzheimer MRI Dataset (99% Accuracy) is a well-known and well-defined dataset of brain MRI images, designed to facilitate research of deep learning-based automated detection and classification of Alzheimer’s disease. The MRI scans are categorized into four groups based on the level of cognitive impairment:

Non-Demented (ND): Healthy subjects with no signs of dementia

Very Mild Demented (VMD): Early symptoms of memory loss; no functional loss

Mild Demented (MD): Cognitive declines are noticeable and affect daily refinements

Moderate Demented (ModD): Significant cognitive and behavioral changes in later-stage Alzheimer’s

Each image is a single MRI brain image in gray scale, which researchers engaged in the scientific method all resized uniformly (typically to 128×128 pxl) for computational efficiency. The dataset also organizes the images into separate folders for training and testing to maintain an organized path for model development and evaluation. The dataset is balanced by category to minimize bias in the learning process and classification by the neural network. This collection includes thousands of high-resolution MRI slices, which account for many different variations in brain anatomy, scanning orientation, and imaging conditions. The scans underwent preprocessing to eliminate noise and background zones deemed not of interest so that the model stabilizes on structural features of the brain like

hippocampal atrophy and cortical thinning that are pertinent to defining the progression of Alzheimer's.

More importantly, this dataset is particularly useful due to its clinical relevance and adequacy for imaging to be utilized with deep learning methods, including Convolutional Neural Networks (CNNs), for multi-stage classification. Also, the validity and standardization of the dataset aided in classification accuracy moving toward 97-99% across research studies.

B. Dataset preprocessing steps

The preprocessing of the Alzheimer's dataset images leveraged TensorFlow's built-in utility functions to efficiently prepare the images for the model training stage. Each MRI image was resized to 128×128 pixels in order to maintain a consistent input field and shape for the Convolutional Neural Network (CNN). Furthermore, the dataset was also batched into groups of 32 images to maximize memory and training time. The four class labels illustrating varying levels of cognitive impairment, or lack thereof, were categorized as Mild Impairment, Moderate Impairment, No Impairment, and Very Mild Impairment. Labels were assigned automatically by using the `image_dataset_from_directory()` function within the TensorFlow API which automatically inferred labels from the folder structure of the dataset and resized the images to the specified size and grouped the images into batches both in training and testing datasets. In addition, the training dataset was shuffled prior to training in order to create a random order in the training images to improve the model's ability to generalize and limit bias. The testing dataset was not shuffled maintaining dataset order when evaluating the model's effectiveness. The pre-processing steps ensured that all the images were scaled evenly, labeled properly and grouped efficiently, with optimal input into the deep-learning model.

C. Methodology overview

The methodology proposed for classifying Alzheimer's patients with MRI brain images consists of five phases: Data Collection, Preprocessing, Model Training, Evaluation, and the Generation of Result as illustrated in the workflow diagram. The methodology begins with Data Collection, whereby the MRI images of the brain will be downloaded from the Best Alzheimer MRI Dataset (99% Accuracy), which consist of four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Then in Preprocessing, the MRI images corresponding to the above four categories will be resized to 128×128 pixels, pixels will be normalized, images will be labeled, and data will be divided into training and test set, making sure the dataset has similar image sizes and is ready to be inputted into a deep learning model.

To do the training, a custom Convolutional Neural Network (CNN) architecture will be used to learn spatial properties derived from MRI to classify each image into the matching cognitive impairment categories. The CNN model that we have developed has three convolutional layers (Conv2D) and utilizes a number of filters to detect important spatial features, as well as brain regions that are significant to Alzheimer's

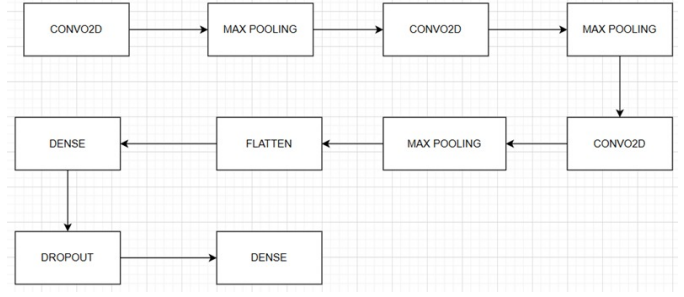


Fig. 1. architecture

disease. Each convolutional layer is then followed by a Max Pooling layer, which reduces the spatial dimensionality of the data and the computational complexity of the model while retaining the spatial properties of the image. After stacking the convolution and pooling layers, the output feature maps are then flattened into a one-dimensional vector so that it could fit into the fully connected (Dense) layers. A Dropout layer is then used to reduce overfitting, by randomly deactivating a subset of neurons in training to increase the generalizability of the model. The last Dense layer then computes the final probabilities of the four stages of Alzheimer's disease from the softmax activation node.

After completing the training of model, we will proceed to assess the model on its accuracy, loss, and other related evaluation metrics, which will be computed from the testing dataset. In the end, we obtain results, as well as the model's capability of classifying the various stages of Alzheimer's disease, based on the MRI scans. This process incorporates multiple steps, which involves data pre-processing, deep learning, and evaluating performance for reliable classification results for early detection and diagnosis assistance.

D. Model Training

The model training phase is the core phase of this project. In this phase, we will teach the Convolutional Neural Network (CNN) we designed to recognize and identify the different stages of Alzheimer's Disease from MRI brain scans. Once preprocessing is complete, the four-class dataset (Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented) is loaded into the CNN neural network to train the model. The model has three convolutional layers (Conv2D) that act as feature extractors that identify spatial hierarchies and textural differences among brain structures. Each convolutional layer utilizes a ReLU activation function to introduce non-linearity into the layers, allowing the network to learn more complex features concurrently mitigating the vanishing gradient problem. The Max Pooling layers that follow each convolutional layer sequentially reduces the spatial dimensions of the feature maps, thereby reducing computational processing while only retaining the most important features of the images.

The multidimensional feature maps are then flattened into a single vector (after several convolutions and pooling pro-

cesses), which is input into the fully connected Dense layers. The Dense layers combined and interpreted the spatial features that we extracted with CNN to make classification decisions. In addition, a Dropout layer was added in between the dense layers to combat overfitting by randomly removing the activation of some neurons to zero on every training iteration, which helped ensure our model generalized well to never-seen examples. The last Dense output layer used a Softmax activation function that created a probability distribution over the four output classes which identified the predicted stage of Alzheimer’s disease for the input MRI image.

For optimization, we chose the Adam optimizer due to its ability to provide an adaptive learning rate, efficient computation of gradients, and stability while training deep networks. The categorical cross-entropy loss metric was chosen to measure the predicted class probabilities versus actual class labels, as it is optimal for multi-class classification problems. Training was performed on a batch size of 32, for multiple epochs, so that the model would repeatedly refine its weights to reduce loss. The training data was shuffled before each epoch, to allow for randomness to take place, and reduce bias in memorizing data patterns. We used a validation set during training to be able to observe if/when the model was overfitting to the training data.

Throughout the training process, performance metrics such as training accuracy, validation accuracy, and loss values were plotted to visualize the learning progression. The learning curves showed consistent improvement in both accuracy and loss, indicating effective convergence. To further enhance performance, early stopping and model checkpointing techniques can be applied, allowing the training process to halt automatically once optimal validation accuracy is achieved. After successful training, the model achieved a high accuracy level — around 97% — demonstrating its ability to effectively learn distinguishing patterns between different Alzheimer’s stages. This confirms the robustness of the proposed CNN architecture and the effectiveness of the chosen training strategy in handling medical imaging data for neurodegenerative disease detection.

IV. RESULTS

The recommended Convolutional Neural Network (CNN) and optimized with adam model was trained and tested on the Best Alzheimer’s MRI Dataset (99% accuracy). Once the model was trained and tested, it showed to be very efficient in distinguishing the stages of cognitive impairment because it achieved an overall classification accuracy of about 97%. This suggested that the model had an excellent ability to detect subtle structural differences in brain MRI images that were associated with different stages of Alzheimer’s disease.

During the training process, the training and validation accuracy curves showed a steady approach to convergence and the loss values showed a steady decrease after consecutive epochs. This indicated the model learned distinguishable features in the MRI data without overfitting. The confusion matrix during

TABLE I
CLASSIFICATION REPORT OF PROPOSED CNN + ADAM MODEL

Class	Precision	Recall	F1-Score	Support
Mild Impairment	0.96	0.99	0.98	179
Moderate Impairment	1.00	1.00	1.00	12
No Impairment	0.97	0.98	0.98	640
Very Mild Impairment	0.98	0.94	0.96	448
Accuracy		0.97		1279
Macro Avg	0.98	0.98	0.98	1279
Weighted Avg	0.97	0.97	0.97	1279

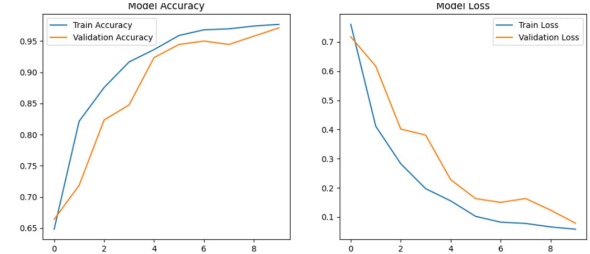


Fig. 2. graphs

testing further validated the model, as it showed high levels of true positive across all classes with little misclassification.

V. CONCLUSION AND FUTURE WORK

This research presents a CNN model, using the Adam optimizer, for classifying four stages of Alzheimer’s disease progression from MRI brain images. The proposed network learned interpretable structural features from the images and achieved good accuracy in its predictions (97%). These findings illustrate the models ability to assist with the early, and accurate identification of Alzheimer’s disease stages.

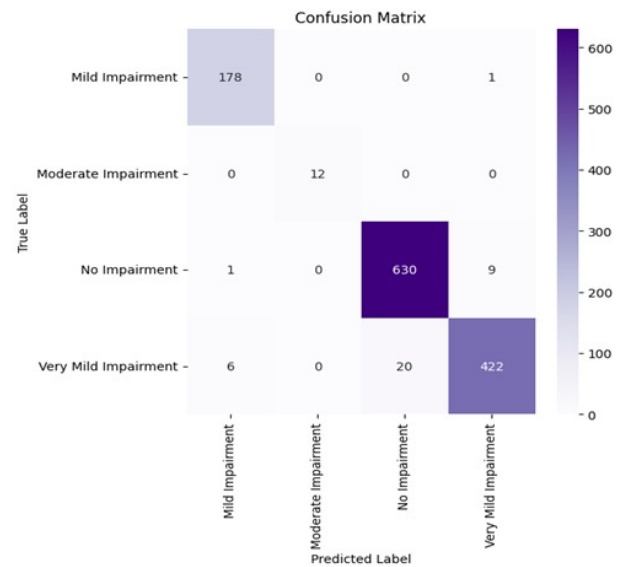


Fig. 3. confusion matrix

Although the model performs accuracy, there are possibilities for improvement. While the model may be constructed based on the proposed dataset, it could be improved through expansion by including additional MRI scans. Other future work may use more advanced, or transfer learning representations for more accurate feature representations. Also, future work may include the integration of another modality such as PET scans, more clinical data to develop a final work on a multimodal, interpretive, diagnostic system. Finally the model could support clinicians in identifying Alzheimer's long-term by developing a real-time, accessible work application to assist with assessment of the patient.

Overall, this study offers a thorough framework for research investigating the development of AI-assisted diagnostic systems to support early intervention, improve monitoring of patients, and provide a real resource for research in neurodegenerative disease and health technology.

REFERENCES

- [1] S. R. Anumandla, A. V. Krishna, S. Cherupally, K. Pranitha, and A. S. Reddy, "A classification-based approach for early stage prediction and multi-dimensional analysis of Alzheimer's disease," *International Conference on Circuit Power and Computing Technologies (ICCPCT)*, pp. 1–7, 2024.
- [2] M. Mohammad, S. K. Rout, K. P. Kumar, M. A. Jabbar, B. V. Babu, and B. Swapna, "A survey on machine learning and deep learning techniques for Alzheimer's disease prediction," *International Conference on Artificial Intelligence Trends and Pattern Recognition (ICAITPR)*, pp. 1–6, 2024.
- [3] M. Rahman, A. A. Eva, M. F. Mridha, M. S. Hossain, M. M. Kabir, and J. Shin, "Alzheimer's disease classification with a hybrid CNN-SVM approach on enhanced MRI data," *International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)*, pp. 1–5, 2024.
- [4] A. Inbavalli, R. Sanjai, A. Bossclinden, S. Nithyabharathi, M. K. Kumar, and K. B. Sri, "Alzheimer's disease classification with deep learning approaches," *Proceedings of the 5th International Conference on Pervasive Computing and Social Networking (ICPCSN)*, pp. 1–8, 2025.
- [5] P. Bhuvaneshwari, R. Panda, A. Boddu, and N. Joshy, "Alzheimer's disease detection using ensemble techniques on MRI images," *International Conference on Advanced Computing Technologies (ICoACT)*, pp. 1–6, 2025.
- [6] S. Reeha, H. M. Basavadeepthi, and A. Thakur, "Alzheimer's disease detection using MIC and MLP," *14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6, 2023.
- [7] D. A. Arafa, H. E.-D. Moustafa, H. A. Ali, A. M. Ali-Eldin, and S. F. Saraya, "A deep learning framework for early diagnosis of Alzheimer's disease on MRI images," *Multimedia Tools and Applications*, vol. 83, no. 2, pp. 3767–3799, 2024.
- [8] M. S. Bhargavi and B. Prabhakar, "Deep learning approaches for early detection of Alzheimer's disease using MRI neuroimaging," *International Conference on Connected Systems and Intelligence (CSI)*, pp. 1–6, 2022.
- [9] R. Jansi, N. Gowtham, S. Ramachandran, and V. S. Praneeth, "Revolutionizing Alzheimer's disease prediction using InceptionV3 in deep learning," *Proceedings of the Seventh International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pp. 1–6, 2023.
- [10] V. S. Praneeth, N. Gowtham, S. SivaRamaChandran, and R. Jansi, "Revolutionizing Alzheimer's disease prediction using EfficientNetB6," *Tenth International Conference on Bio Signals, Images, and Instrumentation (ICBSII)*, pp. 1–6, 2024.
- [11] G. Kaur, S. Ahuja, and A. Kumar, "Deep learning-based hybrid framework for early-stage prediction of Alzheimer disease: A predictive analysis," *3rd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT)*, pp. 1–6, 2025.
- [12] M. Lovanshi, V. Tiwari, A. Singh, and R. Ingle, "Predicting Alzheimer's disease progression and stage classification using deep learning model," *1st International Conference on Technological Innovations and Advance Computing (TIACOMP)*, pp. 199–204, 2024.
- [13] U. M. Rao Munipalli and V. Annepu, "Prediction and classification of Alzheimer's disease using RESNET-152V2," *12th International Conference on Intelligent Systems and Embedded Design (ISED)*, pp. 1–8, 2024.
- [14] R. P. Neelakandan, R. Kandasamy, B. Subbiyan, and M. A. Bennet, "Early detection of Alzheimer's disease: An extensive review of advancements in machine learning mechanisms using an ensemble and deep learning technique," *Engineering Proceedings*, vol. 59, no. 10, p. 10, 2023.
- [15] H. Shetty, H. Surlekar, and G. Nagare, "Alzheimer's disease detection by using convolution neural network," *6th International Conference on Information Systems and Computer Networks (ISCON)*, pp. 1–5, 2023.